

10. (a) The mass spectrum of a silicon sample showed the presence of the three isotopes ^{28}Si , ^{29}Si and ^{30}Si . The percentage of ^{28}Si present was 92.2% and the mass spectrum showed that the percentage of ^{29}Si present was twice that of ^{30}Si .

Calculate the relative atomic mass of this sample of silicon.

[3]

You **must** show your working.

Relative atomic mass =

- (b) Rhiannon studied the properties of silicon and found that its structure is similar to that of diamond. In her report she stated that

- the Si–Si–Si bond angle is 109.5°
- silicon is a poor conductor of electricity
- each silicon atom is bonded to four other silicon atoms.

- (i) State the name of the shape that has this bond angle.

[1]

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- (ii) Explain why solid silicon is a very poor electrical conductor.

[1]

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- (iii) Explain why the bonding between each atom is covalent.

[1]

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- (c) Hydrofluoric acid, HF, is supplied as a solution that contains 50 % by mass of hydrogen fluoride. The density of this solution is 1.17 g cm^{-3} .

Calculate the concentration of this solution in mol dm^{-3} . [2]

Concentration = mol dm^{-3}

- (d) (i) Hydrofluoric acid will dissolve silica, SiO_2 , to produce hexafluorosilicic acid. This acid contains the SiF_6^{2-} ion.

Use the information from the table below to draw the shape of the SiF_6^{2-} ion, showing the F–Si–F bond angle. Give a reason for your answer. [3]

Ion	Number of bonding electron pairs	Number of lone electron pairs on the central silicon atom
SiF_6^{2-}	6	0

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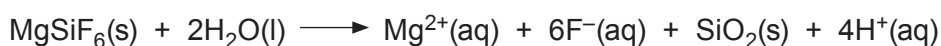
- (ii) Hexafluorosilicic acid, H_2SiF_6 , (M_r 144) can be added to drinking water to promote good dental health. When added to water all the fluorine in the acid is available as fluoride ions.

The water in an area of Derbyshire contains 0.15 mg dm^{-3} of fluoride ions.

Calculate how much hexafluorosilicic acid, in mg, should be added to each dm^3 of water to increase the fluoride level to 0.76 mg dm^{-3} . [3]

Mass of hexafluorosilicic acid = mg

- (e) Solid magnesium hexafluorosilicate hydrolyses rapidly when added to water. One equation for this reaction is as follows.



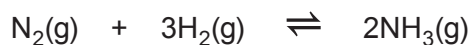
In an experiment 2.60 g of magnesium hexafluorosilicate were added to 1.00 dm^3 of water.

Calculate the pH of the resulting mixture. [3]

pH =



10. (a) Nitrogen can react with hydrogen to form ammonia.



A mixture of N_2 and H_2 is left to react at a certain temperature, until it reaches equilibrium. The equilibrium mixture has the following composition.

N_2	1.16 mol dm^{-3}
H_2	1.60 mol dm^{-3}
NH_3	$0.752 \text{ mol dm}^{-3}$

- (i) A student said that the equilibrium must lie to the left because the concentrations of nitrogen and hydrogen are greater than that of ammonia. Is he correct?

Justify your answer by calculating a value for K_c for this equilibrium.
Give the unit for K_c .

[4]

 $K_c = \dots\dots\dots$ Unit $\dots\dots\dots$

$\dots\dots\dots$
 $\dots\dots\dots$

- (ii) When the temperature is increased the equilibrium yield of NH_3 decreases. The student said that the reaction is endothermic. Is he correct?

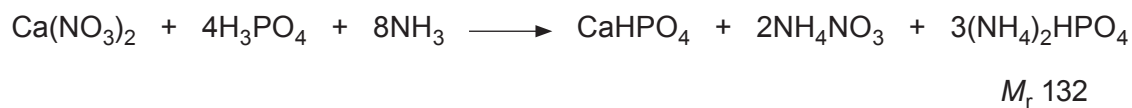
Justify your answer by using Le Chatelier's principle.

[2]

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 $\dots\dots\dots$
 $\dots\dots\dots$



- (b) Ammonia can be used as part of the nitrophosphate process to produce the fertiliser diammonium hydrogenphosphate (DAP) which has the formula $(\text{NH}_4)_2\text{HPO}_4$.



Calculate the maximum mass of DAP, in kg, that could be made from 1.00 tonne of ammonia. [3]

Maximum mass = kg

- (c) Calculate the volume, in cm^3 , that 2.54×10^{-3} mol of nitrogen occupies at a temperature of 120°C and a pressure of 101 kPa. [4]

Volume = cm^3



10. When an electrical discharge passes through gaseous hydrogen at low pressure, electromagnetic radiation is emitted.

- (a) Describe the processes within a hydrogen atom that cause electromagnetic radiation to be emitted. [2]

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- (b) If the electromagnetic radiation in part (a) is passed through a spectrometer, several series of converging lines are observed.

- (i) Explain why there are several series of lines. [1]

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- (ii) Explain why the lines within each series converge. [1]

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- (c) The convergence limit of the Lyman series of lines occurs at a wavelength of 91.2 nm.

- (i) State what the limit represents. [1]

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- (ii) Calculate the energy, in kJ mol^{-1} , of the convergence limit. [4]

Energy = kJ mol^{-1}



- (d) Hydrogen forms when aluminium reacts with sulfuric acid.



- (i) Calculate the volume of hydrogen, in cm^3 , that would be produced if 0.131 g of aluminium were added to an excess of sulfuric acid at a temperature of 25°C and a pressure of 1 atm. [3]

Volume = cm^3

- (ii) Calculate the volume of hydrogen that would be produced if the same experiment were carried out at 50°C and 1.6 atm. [2]

(If you do not have an answer in part (i), assume that the volume is 200 cm^3 . This is **not** the correct answer.)

Volume = cm^3



10. (a) Group 1 and Group 2 metals are in the s-block of the Periodic Table. Using potassium and calcium as examples, discuss the similarities and differences between Group 1 and Group 2 with respect to:

- the reaction of the metals with cold water.
- the solubility of the carbonates.

[6 QER]

You should include appropriate chemical equations in your answer.

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- (b) The equation for the reaction between potassium carbonate and hydrochloric acid is given below.



A 1.40 g sample of impure potassium carbonate was added to excess dilute hydrochloric acid. The impurity is unreactive and only the potassium carbonate reacts with the acid.

The volume of carbon dioxide released was 186 cm^3 when measured at 298 K and $1.01 \times 10^5\text{ Pa}$.

Calculate the mass of the impurity.

[3]

mass of impurity = g

- (c) A solution is thought to contain potassium chloride. Describe suitable tests that a student could do to confirm this. Include the expected observations.

[2]

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- (d) By referring to ionisation energies, explain why stable compounds containing K^{2+} ions are unlikely to form.

[2]

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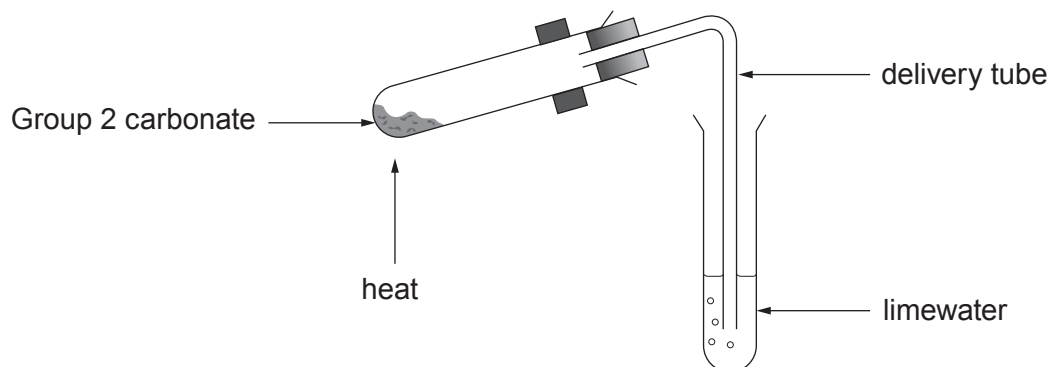
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10. (a) A student investigated the thermal stability of Group 2 carbonates. She used the following apparatus and method.



1. Start a stopwatch at the moment you begin to heat the carbonate and continue to heat for 4 minutes or until the limewater turns cloudy.
2. After the limewater turns cloudy lift the delivery tube out of the limewater then remove the flame from under the boiling tube.
3. Repeat the heating procedure for each carbonate in turn.
4. Use a spatula-measure of the appropriate carbonate, fresh limewater and heat with the hottest Bunsen burner flame each time.

Her results are shown below.

Carbonate	Time taken for limewater to turn cloudy/s
MgCO ₃	20
CaCO ₃	40
SrCO ₃	230
BaCO ₃	does not turn cloudy



- (i) Suggest an improvement to the method to ensure that the experiment is a fair test. [1]

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- (ii) Suggest why the delivery tube should be lifted out of the limewater before the flame is removed from under the boiling tube. [1]

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- (iii) State what conclusion she can draw about the thermal stabilities of the Group 2 carbonates from these results. [1]

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- (iv) The student was told that the temperature at which barium carbonate decomposes is 1360°C . The maximum temperature of a typical Bunsen burner flame is around 800°C .

State whether the limewater would have turned cloudy if she had used two Bunsen burners to heat the barium carbonate. Give a reason for your answer. [1]

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- (b) Barium nitrate also decomposes on heating.



- (i) In an experiment 0.960 g of barium nitrate was heated strongly for 2 minutes.

Calculate the maximum volume, in cm^3 , of gas that could be produced at a temperature of 25°C and a pressure of 1 atm.

[3]

Volume = cm^3

- (ii) The volume of a gas is directly proportional to its temperature at constant pressure.

A student said that if the gas formed in this experiment were collected at a temperature of 50°C and at 1 atm pressure, the volume formed would be double that calculated in part (i).

Do you agree? Justify your answer.

[1]

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- (c) State the conditions necessary for **each** of barium oxide and barium metal to conduct electricity. Explain this property in terms of structure and bonding in each case. [3]

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- (d) The atomic radius of a barium atom is 0.217 nm.

From the list below, choose the value for the **ionic** radius of a barium ion. Give a reason for your choice. [2]

0.135 nm

0.210 nm

0.217 nm

0.265 nm

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- (e) A sample of barium contains two isotopes. The first isotope has a relative isotopic mass of 134.9 and the second a relative isotopic mass of 137.9. The relative atomic mass of the sample of barium is 137.3.

Calculate the percentage abundance of the first isotope. [2]

Abundance = %

15

